Guide to Part 2

Purifying a Glowing Protein · BCH 3023 · measuring what you purified: the Bradford assay, the fluorimeter and the gel

Part 2 measures the samples you kept in Part 1: how much protein each holds, how brightly it glows, what its excitation and emission spectra look like, and how big it is on a gel. You then put it all together in a purification table and decide which protein and which tag your unknown is. This guide takes you through every step: **what to do**, **what you should see**, and **why** the step matters at a real bench.

Before you start

- **Finish Part 1 first,** on the same computer and browser: Part 2 works on the samples it kept. Tick the elution fractions to pool in the Pool column of Table 1 (Part 1) before you start.
- Open Part 2 from the link under the title of Part 1 (or add `?part=2` to the address). Your code is the same.
- Allow about two hours: the assay and the fluorimeter take about 45 minutes, the gel about an hour and a quarter. `Speed up the wait` runs the bench clock 60 times faster during the waits.
- Every step can be done in the 3D room (click a thing to pick it up, then click where it goes) or with the buttons beside the room. This guide names the buttons.
- The scans and the gel pictured in this guide are of a **made-up practice protein, Lanternin** (the one in the prelab), not of any of the unknowns.

At a real bench

- **Acrylamide is a nerve poison** until it has set into a gel. Wear gloves when you make up and pour the gel mixtures, and clean up drips at once.
- **The gel tank carries a dangerous current.** Put the lid on before switching the power supply on, and switch it off before you lift the lid. (The power supply here will not start with the lid off.)
- UV light harms eyes and skin: never look into a UV lamp, and close the imager's door before switching a light on.
- Coomassie stains skin and the destain holds methanol and acetic acid: gloves, and work over the tray.
- Bradford dye is acidic (it holds phosphoric acid): wipe up spills.

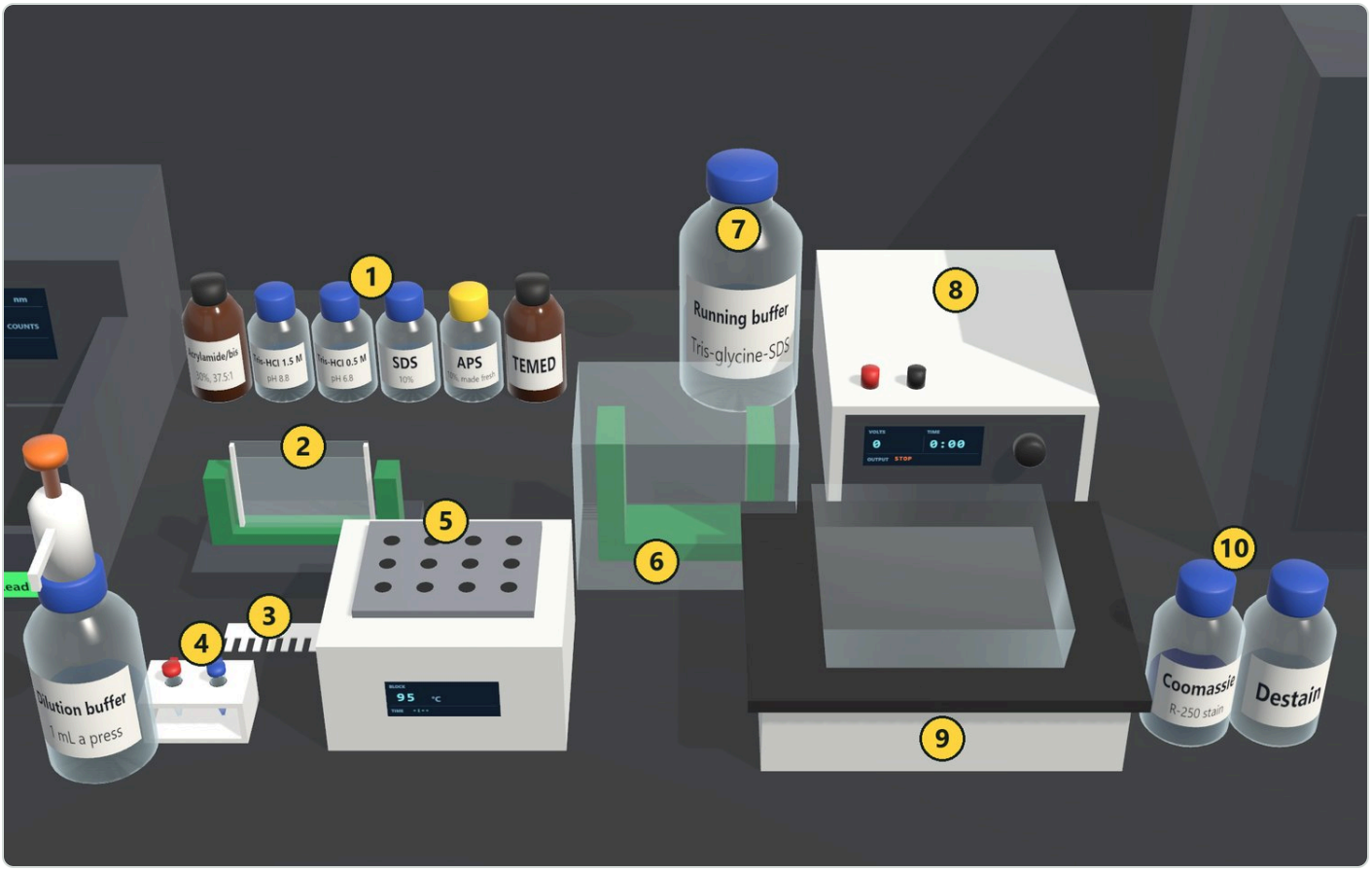
THE BENCH

Where everything is



The meters (the Meters view), with the blank and the standards already made up in their cuvettes: brown with no protein, bluer with more.

1. The arm's micropipette
2. Bradford dye, 1 mL a press
3. The seven BSA standards
4. Thirteen cuvettes: blank, S1 to S7, five samples
5. Spectrophotometer (cut open)
6. Your samples: lysate, flow-through, washes
7. Fractions, the pool and the dilutions
8. Fluorimeter (cut open)
9. The quartz cuvette
10. Dilution buffer, 1 mL a press



The gel bench (the Gel bench view).

1. Gel reagents
2. Casting stand with its plates
3. The comb
4. Laemmli buffer and the ladder
5. Heat block (95 °C)
6. Gel tank
7. Running buffer
8. Power supply
9. Rocker with the staining tray
10. Coomassie and destain

The views under the room (Room, Meters, Spectrophotometer, Fluorimeter, Gel bench, Casting, Gel tank, Imager) move the camera; most buttons move it for you. The micropipette's volume (under the room) is used by every pipetful: set it to **20 µL** for the assay and the fluorimeter, and to **10 µL** to load the gel. Each step beside the room has a light: a blue ring with **Next** on the step to do now, a green tick when it is done.

PART A

Your samples: pool and dilute

1 Pool the fractions you chose

Press Pool them. The elution fractions you ticked in Table 1 of Part 1 are poured together into the Pool tube.

The step says how much was pooled and from which fractions. Table 4 shows the pool's volume: you need it for the purification table.

Why: the fractions that glow hold your purified protein; together they are one sample to measure.

2 Dilute the samples that need it

For each sample, choose a dilution (1:2 to 1:100) and press Make it. A dilution is made in dilution buffer, 200 µL in all: 1:10 is 20 µL of sample and 180 µL of buffer.

Why: the Bradford assay is close to a straight line only up to about 1 mg/mL, and the dye tolerates no more than 0.2 M imidazole (the elution buffer has 250 mM). A cleared lysate often holds 5 to 10 mg/mL of protein; a purified pool 1 to 5. Too dilute is a problem too: a reading below the lowest standard is not reliable.

Write each dilution down. Every concentration you work out from a diluted sample must be multiplied by its dilution factor. If a reading comes out off the straight part of the curve, make a stronger dilution and measure again: the page shows which dilution each cuvette got.

PART B

How much protein: the Bradford assay

The Bradford dye is brown; when it binds protein it turns blue, and the blue absorbs at 595 nm. Seven BSA standards of known concentration make a standard curve, and each sample is read against it.

3 1 mL of dye into each of the thirteen cuvettes

Choose a cuvette in the **Cuvette** box and press 1 mL of dye; repeat for all thirteen. In the room: click the dye bottle, then each cuvette (the dispenser stays in your hand).

4 20 µL of its own sample into each cuvette

Set the micropipette to 20 µL. Choose each cuvette and press Add its sample: the blank gets dilution buffer, S1 to S7 their BSA standards, and each sample's cuvette its sample (its dilution, if you made one). Empty it starts a cuvette again.

Why the blank: dye with no protein. Zeroing on it means every reading after it is the blue the protein made, not the dye's own colour.

5 Wait at least 5 minutes, and read within the hour

The step shows how long the last cuvette still needs. Speed up the wait helps.

Why: the blue takes a few minutes to develop; read too soon and the reading is low. After an hour the dye and the protein start to come out of solution and the reading drifts.



Step 6: the blank in, zeroed: 0.000. The orange beam goes in at 595 nm and comes out to the detector.



Step 7: standard S5 (1 mg/mL). Its blue absorbs some of the beam: the beam leaving the cuvette is weaker.

6 Zero the spectrophotometer on the blank

Press Blank in, then Zero. The screen reads 0.000 and **BLANK**.

7 Read every cuvette and record it

Choose a cuvette and press Into the spectrophotometer (the one already in it comes out). The screen shows its absorbance at once. Type it, exactly as shown, into Table 3 (the standards) or Table 4 (your samples).

> A reading that matches the screen gets a green tick. The page draws your standard curve (Figure 1) as you go and fits a straight line through the blank and the standards up to 1 mg/mL.

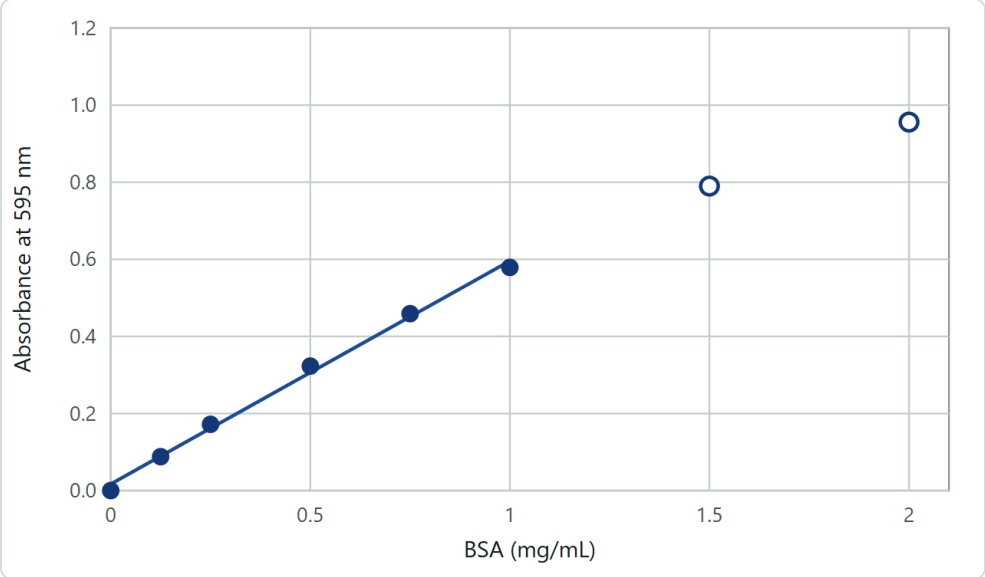


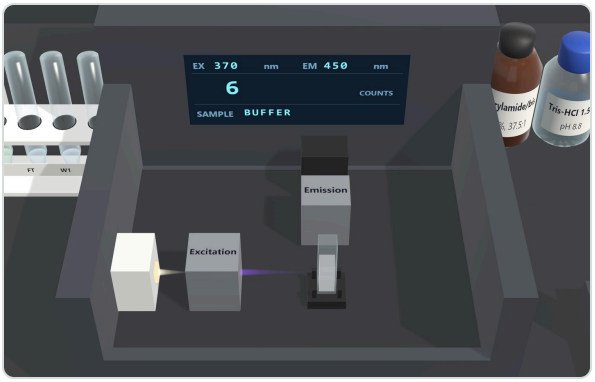
Figure 1, drawn by the page from Table 3: the line through the blank and the standards up to 1 mg/mL (filled points). Above 1 mg/mL (open circles) the curve bends over, which is why a sample must read on the straight part.

Working out a sample (made-up numbers). The line is A_{595} = 0.579 × (mg/mL) + 0.017. A sample diluted 1:10 reads 0.412. In the cuvette's sample (the dilution): (0.412 − 0.017) ÷ 0.579 = **0.682 mg/mL** (the "From the curve" column). In the sample itself: 0.682 × 10 = **6.82 mg/mL** (the "In the sample" column).

PART C

How brightly it glows: the fluorimeter

A fluorescent protein takes up light of one colour (excitation) and gives off light of a longer wavelength (emission). The fluorimeter shines one wavelength in and, at a right angle to the beam, counts the light that comes out at a second wavelength. Its detector counts up to 10,000: above that the screen reads **OVER**.



The quartz cuvette in the fluorimeter, with buffer only: excited at 370 nm (the violet beam), it reads just 6 counts. A glowing sample sends its own light out at a right angle, toward the emission monochromator.

8 Fill the quartz cuvette: buffer first

Press [1 mL of buffer], choose a sample in the **Sample** box, and press [20 µL of the sample] (the micropipette at 20 µL; its dilution is used if you made one). Then [Put it in]. Between samples, [Take it out] and [Empty and rinse].

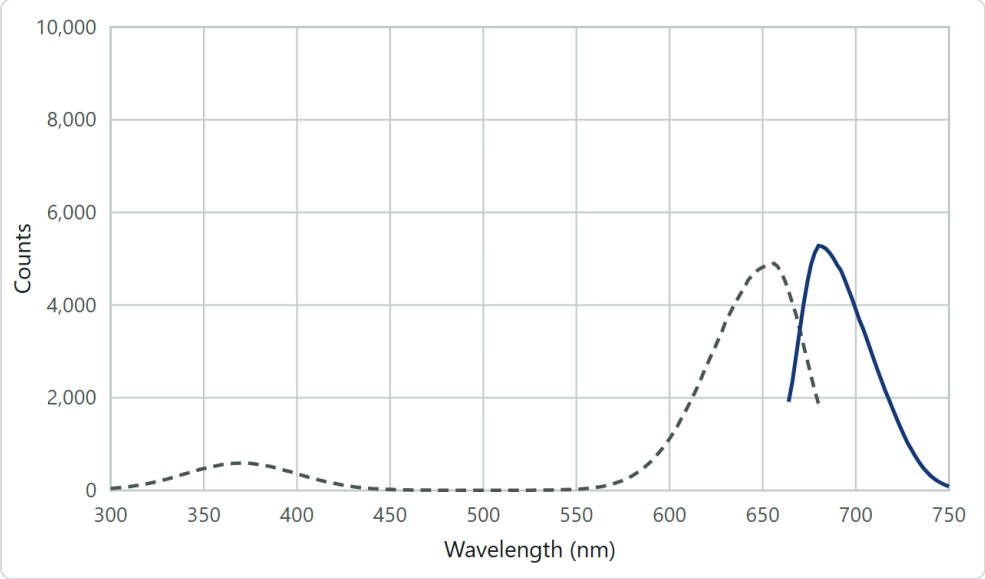
> **Why buffer first:** the beam crosses the cuvette about 1 cm up. 20 µL alone would leave the beam passing over the liquid.

9 Find the peaks: three scans of the pool

1. **Emission scan, exciting at 370 nm.** Type 370 in **Excitation**, press [Set], then [Emission scan]. These proteins all take up a little light in the near ultraviolet, so this shows where the emission is, whatever the protein.
2. **Excitation scan.** Set **Emission** to the peak you found (or a little past it) and press [Excitation scan].
3. **Emission scan again,** exciting at the excitation peak: the clearest emission peak.

> Figure 2 draws your latest excitation scan (dashed) and emission scan (solid). Point at a curve to read its wavelength and counts. Type the two peaks into the boxes under the figure: the page checks them against your own scans.

> **A flat top means OVER:** the detector was full and the true peak is cut off. Make a stronger dilution of the pool and scan again. An emission scan starts 10 nm above the excitation wavelength; light scattered by the cuvette makes its first few points rise.



The practice protein Lanternin (made up). Dashed: the excitation scan, watching at 690 nm, peaks at 655 nm (the small hump near 370 nm is the near-ultraviolet band). Solid: the emission scan, excited at 654 nm, peaks at 680 nm.

10 Read every sample at the two peaks

Set **Excitation** and **Emission** to your peaks and press Set. Then, for each sample: fill the quartz cuvette, put it in, press Read, and record the counts in Table 5. The fluorescence per mL is the reading times the dilution.

> Read **every** sample at the **same** two wavelengths, or the readings cannot be compared. A reading of OVER: dilute that sample more.

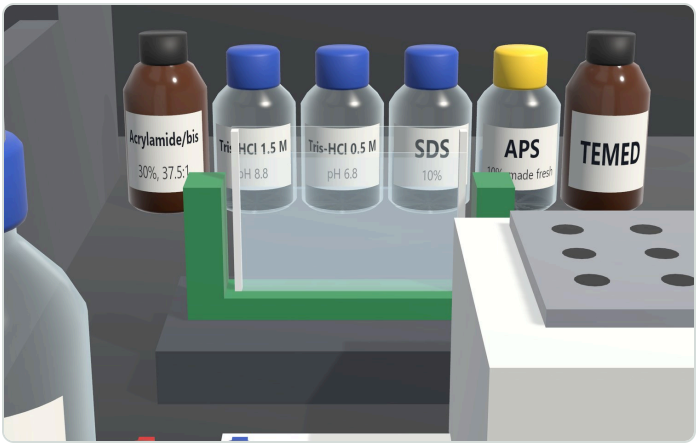
PART D

The gel

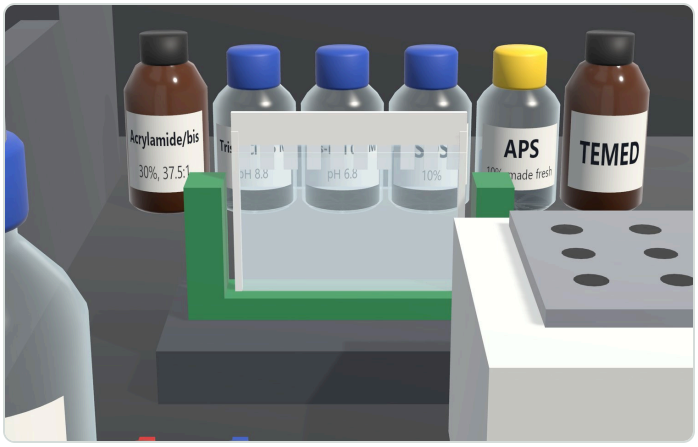
SDS-PAGE sorts proteins by size. SDS coats every protein with negative charge and heat unfolds it, so in an electric field all of them run toward the positive electrode, and the gel's mesh holds the big ones back.

Reagent	Resolving gel, 10% (10 mL)	Stacking gel, 4% (10 mL)
Water	4.1 mL	6.1 mL
Acrylamide/bis, 30%	3.3 mL	1.3 mL
Tris-HCl 1.5 M, pH 8.8	2.5 mL	–
Tris-HCl 0.5 M, pH 6.8	–	2.5 mL
SDS, 10%	0.1 mL	0.1 mL
APS, 10% (last but one)	50 µL	50 µL
TEMED (last)	5 µL	10 µL

Recipes from Bio-Rad's Mini-PROTEAN manual. Each reagent's button adds its volume.



Step 12: the resolving gel poured between the plates and covered with water.



Step 13: the stacking gel poured on top and the comb pushed in.

11 Mix the resolving gel

Press each of its reagent buttons in step G: water, acrylamide/bis, **Tris pH 8.8**, SDS, then APS and TEMED last. Both Tris buffers are offered: choose the right one.

> **Why last:** APS and TEMED together start the acrylamide setting into a gel. Pour within a few minutes, or it sets in the tube.

12 Pour it, cover it with water, and wait about 45 minutes

Press `Pour`, then `Cover with water`. The step counts down until it has set.

> **Why the water:** oxygen in the air stops acrylamide setting. Covered, the top sets flat and even; uncovered, it sets soft and wavy, and so will your bands.

13 The stacking gel and the comb

Mix the stacking gel the same way (with **Tris pH 8.8**), then `Pour` (the water is poured off first) and `Comb in` at once. After about 30 minutes, `Comb out`: ten wells. Then `Into the tank` and `Fill with running buffer`.

> **Why two layers:** the stacking gel's lower pH and looser mesh squeeze each sample into a thin band before it enters the resolving gel, where the proteins separate.

14 Prepare the samples: Laemmli buffer, 95 °C for 5 minutes

Press `Add it` (2x Laemmli into every sample you kept), then `Into the heat block`; after 5 minutes on its screen, `Out`.

> **Why:** SDS and heat unfold every protein so it runs by size alone; the glycerol makes the sample sink into its well; the blue dye shows how far the run has got. An unheated fluorescent protein stays partly folded, still glows, and runs faster than its size.

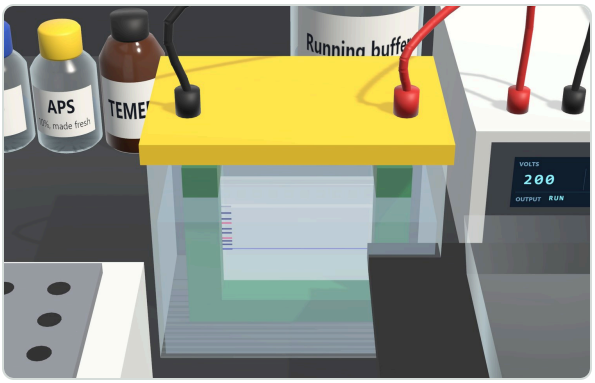
15 Load 10 µL into each well

Set the micropipette to 10 µL. For each well, choose what goes in and press `Load`. Ten wells, eleven samples: put the ladder in one well and choose the nine samples that tell you most (whole cells and the pellet show what was made and lost; the elution fractions show what you purified). Write your plan down.

16 Run at 200 V until the dye front nears the bottom

Press Lid on or off (lid on), Red lead to the red jack, type 200 in **Volts**, and Power on. It takes about 40 minutes (speed it up). When the note says "*The blue dye front is nearly at the bottom of the gel: turn the power off now.*", press Power off.

> **Red to red.** The proteins are negative and run toward the positive (red) electrode at the bottom. With the leads the wrong way round they run up, out of the top of the gel, and are lost. Run too long and the smallest ones run off the bottom.

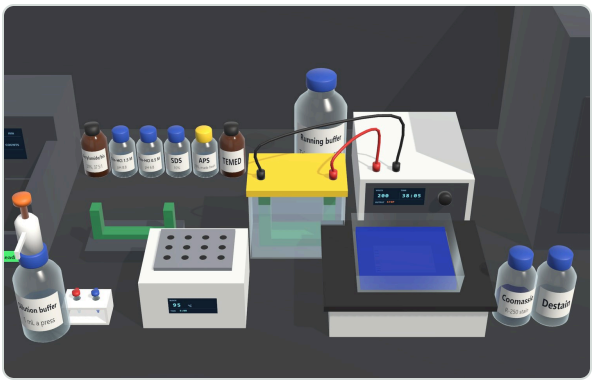


Step 16: running at 200 V. The prestained ladder's bands (blue, two pink) and the blue dye front can be seen moving down; the other proteins cannot until the gel is stained.

17 Stain for an hour, destain for two

Press Into the tray, then Coomassie. After an hour, Destain, and wait about two hours.

> **Why:** Coomassie dyes every protein blue, and the gel too. Destaining takes the blue back out of the gel and leaves the bands. Too short a stain and the faint bands do not show.



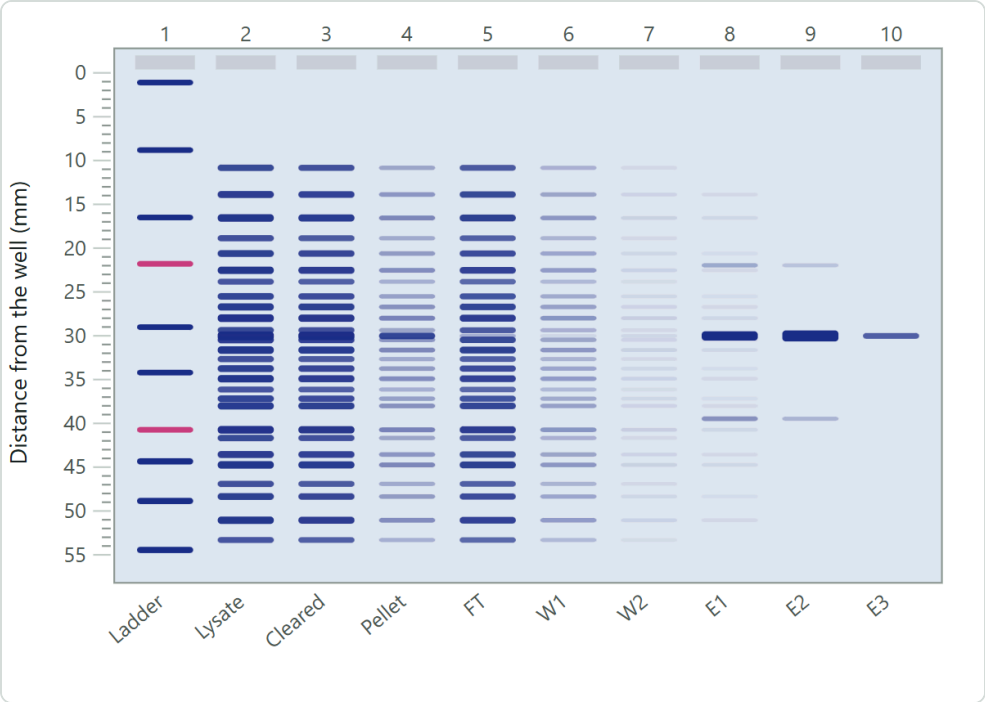
Step 17: the gel in Coomassie on the rocker.

18 Take its picture in the imager

Press Into the imager, White light, then Take the picture. The picture appears in Figure 3. The Blue light shows only what still glows: try it on an unstained gel.

PART E

Reading the gel, and the purification table



A practice gel (made up): the ladder in lane 1 (the pink bands are 75 and 25 kDa), then the lysate, cleared lysate, pellet, flow-through, washes and elution fractions of a made-up His-tagged protein, Lanternin. Its band is the strong one in lanes 8 to 10, at 30.0 mm. The faint bands at about 22 and 39 mm in the elution lanes are two of the bacteria's own proteins that stick to nickel.

19 Measure the ladder

Point at each ladder band in Figure 3: the distance from the bottom of the well to the middle of the band shows beside the gel. Type it into Table 6, with the $\log_{10}$ of each band's mass ($\log_{10}$ 50 = 1.699).

> Figure 4 plots $\log_{10}$(mass) against distance and fits a straight line through the bands from 15 to 100 kDa, where the gel separates by size most evenly.

Band (kDa)	Distance (mm)	$\log_{10}$(mass)
250	1.1	2.398
150	8.8	2.176
100	16.5	2.000
75	21.8	1.875
50	29.0	1.699
37	34.2	1.568
25	40.7	1.398
20	44.3	1.301
15	48.9	1.176
10	54.5	1.000

Table 6 for the practice gel.

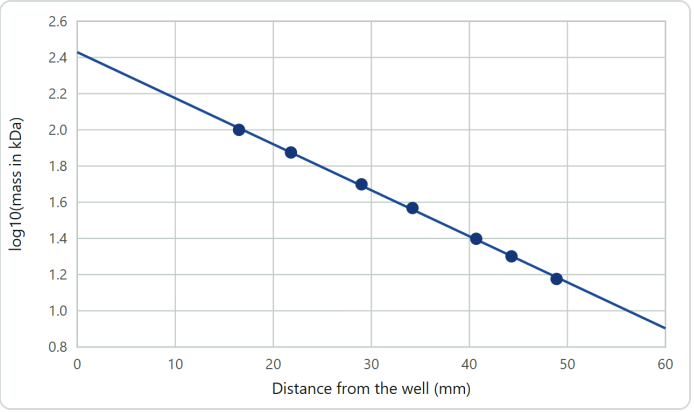


Figure 4: the line through the ladder's bands from 15 to 100 kDa. Above 100 kDa the big proteins barely enter the gel, and below 15 kDa they run close to the dye front, so those bands are left out of the line.

20 Read your protein's mass

In Table 7, choose a lane with your protein (an elution fraction), measure its band, read $\log_{10}$(mass) off the line, and take 10 to that power.

The practice gel. The line is $\log_{10}(\text{mass}) = -0.0254 \times (\text{mm}) + 2.429$. Lanternin's band is at 30.0 mm: $\log_{10}(\text{mass}) = -0.0254 \times 30.0 + 2.429 = 1.667$, so the mass is $10^{1.667}$ = **46.5 kDa**. (Lanternin is 46 kDa alone, about 47 with its His-tag.)

21 The purification table

For each sample in Table 8: **total protein** = concentration (Table 4) × volume; **total fluorescence** = fluorescence per mL (Table 5) × volume; **fluorescence per mg** = total fluorescence ÷ total protein; **yield** = its total fluorescence ÷ the cleared lysate's × 100; **purification** = its fluorescence per mg ÷ the cleared lysate's.

Made-up numbers.

Sample	Volume	mg/mL	Fluorescence per mL	Total protein	Total fluorescence	Per mg	Yield	Purification
Cleared lysate	4.80 mL	8.00	20,000	38.4 mg	96,000	2,500	100%	1
Pool	1.20 mL	2.50	70,000	3.00 mg	84,000	28,000	87.5%	11.2

22 Which protein is it?

Match your two peaks to the reference sheet on the page, and your mass to the protein's mass plus a tag's. Choose the protein and the tag, and write your evidence: which measurements point to them, and how you ruled out the others. The page does not check this; your instructor marks it.

WHEN SOMETHING GOES WRONG

Messages, and what to do

The bench says	What to do
The micropipette is set to … µL	Set it to 20 µL for the assay and the fluorimeter, 10 µL to load the gel.
This sample holds about … mM imidazole	Above 200 mM the dye turns blue by itself. Make a stronger dilution, empty the cuvette, and make it again.
The spectrophotometer has not been zeroed yet	Blank in, then Zero.
This cuvette was mixed less than 5 minutes ago	Wait (or speed up the wait), then read it.
A reading above the straight part of the curve	Make a stronger dilution of that sample and measure it again.
OVER on the fluorimeter, or a flat-topped scan	The detector is full: dilute more and read or scan again.
Your samples were read at different settings	Read every sample at the same excitation and emission wavelengths.
Not that Tris	pH 8.8 for the resolving gel, pH 6.8 for the stacking gel.
APS and TEMED go in last	Add the water and the acrylamide first.
It set in the tube / it will never set	Cast a fresh gel (the note offers it); this time pour within a few minutes, with both APS and TEMED in.
The leads are the wrong way round	Red lead to the red jack.
The power supply will not start with the lid off	Lid on first.
Something does not agree with the screen or with your own numbers	The hint under the box says what to look at: a reading to take again, a dilution factor, a volume, a ratio the wrong way up.

To start again from a clean bench, use Start over under the steps: this erases **both** parts, so use it only if you must.

Quick reference

Samples

1. Pool them (the fractions ticked in Part 1).
2. Dilute: choose 1:2 to 1:100, Make it. Write it down.

Bradford (micropipette 20 µL)

3. Each cuvette: 1 mL of dye, Add its sample.
4. Wait 5 minutes (read within the hour).
5. Blank in, Zero.
6. Each cuvette Into the spectrophotometer; record Tables 3 and 4.
7. Concentration = (A − intercept) ÷ slope; × dilution.

Fluorimeter (micropipette 20 µL)

8. 1 mL of buffer, 20 µL of the sample, Put it in.
9. Pool: emission scan at 370; excitation scan at the emission peak; emission scan at the excitation peak.
10. Set both peaks; Read every sample; OVER = dilute more.

Gel

11. Resolving (Tris 8.8), APS and TEMED last; Pour, Cover with water; 45 min.
12. Stacking (Tris 6.8); Pour, Comb in; 30 min; Comb out.
13. Into the tank, Fill with running buffer.
14. Add it (Laemmli); 95 °C 5 min.
15. Micropipette 10 µL; ladder in one well; Load.
16. Lid on, red to red, 200 V, ~40 min; off when the dye front nears the bottom.
17. Into the tray, Coomassie 1 h, Destain 2 h.
18. Into the imager, White light, Take the picture.

Results

19. Ladder: distance and $\log_{10}$(mass); your band → mass = $10^{\log}$.
20. Total = per mL × volume; yield = ÷ lysate × 100; purification = per mg ÷ lysate's.
21. Peaks + mass → the reference sheet → protein and tag, with your evidence.